6.1 Area Between Curves

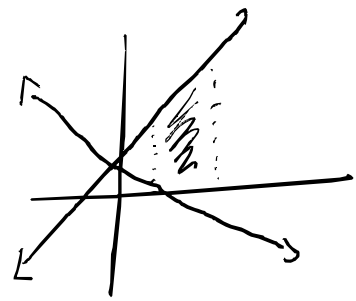
If $y = f(x)$ and $y = g(x)$ are continuous functions and if $f(x) \geq g(x)$ on an interval $a \leq x \leq b$, then the area of the region bounded by the curves on that interval is

$$A = \lim_{n\to\infty} \sum_{i=1}^{n} [f(x_i^*) - g(x_i^*)]\,\Delta x = \int_a^b [f(x) - g(x)]\ dx$$

1. Find the area between the curves $f(x) = 5x + 2$ and $g(x) = -2x + 1$ on the interval from $x = 1$ to $x = 3$.

$A = \int_1^3 (5x+2) - (-2x+1)\,dx$

$A = \int_1^3 7x + 1\,dx$



If $x = f(y)$ and $x = g(y)$ are continuous functions and if $f(y) \geq g(y)$ on an interval $a \leq y \leq b$, then the area of the region bounded by the curves on that interval is

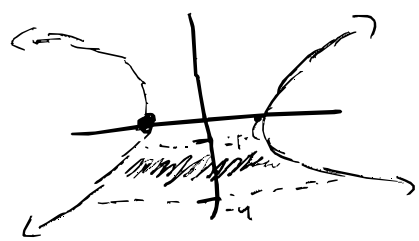
* quadratic transformations

$$A = \int_a^b [f(y) - g(y)]\ dy$$

2. Find the area between the curves $f(y) = y^2 + 1$ and $g(y) = -y^2 - 1$ on the interval from $y = -4$ to $y = -1$.

$A = \int_{-4}^{-1} (y^2+1) - (-y^2-1)$

$= \int_{-4}^{-1} 2y^2 + 2$



3. Find the area of the region(s) bounded by the curves. In most instances in lecture, we will set up the integral, and skip evaluating to save time.

(a) $y = 5x - x^2$, $y = 2x - 4$

$y = x(5-x)$

$x=0, x=5$

bounds: $5x - x^2 = 2x - 4$

$= x^2 - 3x - 4$

$= x^2 + x - 4x - 4$

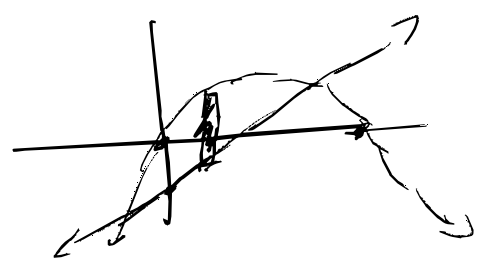
$= x(x+1) - 4(x+1)$

$= (x+1)(x-4)$

$4, -1$

$A = \int_{-1}^{4} (5x - x^2) - (2x - 4)$

$= \int_{-1}^{4} -x^2 + 3x + 4\,dx$

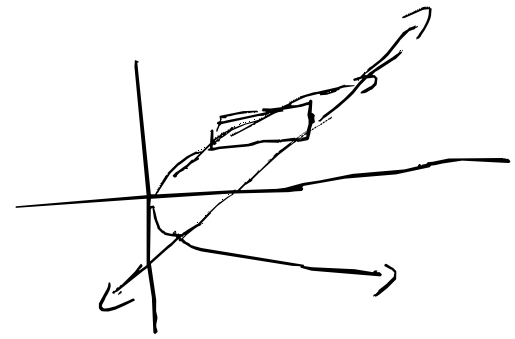


$-2y = 3 - x$

$y = \frac{1}{2}x - \frac{3}{2}$

(b) $x = y^2$, $x - 2y = 3$

$x = y^2 \qquad x = 2y+3$



$A = \int_{-1}^{3} 2y+3-y^2$

$y^2 = 2y+3$

$y^2 - 2y - 3 = 0$

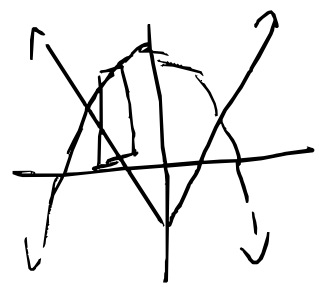
$y^2 + y - 3y - 3 = 0$

$y(y+1) - 3(y+1) = 0$

$(y+1)(y-3) = 0$

$3, -1$

(c) $y = |x| - 1$, $y = 5 - x^2$



$A = 2\int_0^2 (5-x^2) - (x-1)$

$= 2\int_0^2 -x^2 - x + 6$

$5 - x^2 = x - 1$

$= x^2 + x - 6$

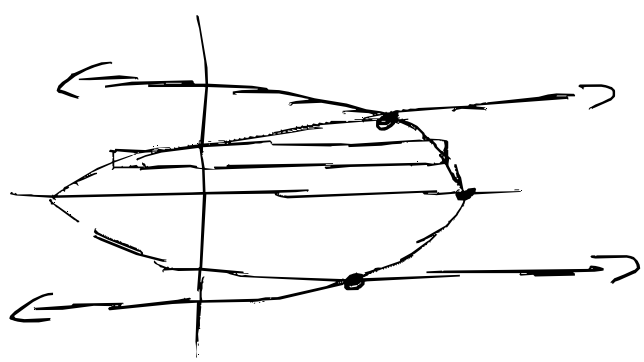
$= x^2 - 2x + 3x - 6$

$= x(x-2) + 3(x-2)$

$= (x+3)(x-2$

$-3, 2$

(d) $x = 10 - y^2, x = 2y^2 - 5$



$A = \int (10-y^2) - (2y^2-5)$

$= \int_{-\sqrt{5}}^{\sqrt{5}} -3y^2 + 15 \; dy$

$10 - y^2 = 2y^2 - 5$

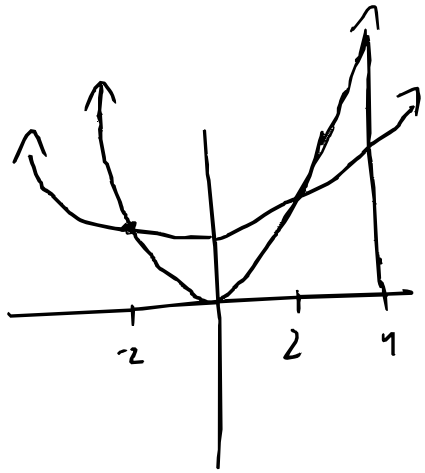
$15 = 3y^2$

$y^2 = 5$

$y = \pm\sqrt{5}$

$\boxed{\text{if } x^2 = 10 \quad x = \pm\sqrt{10}}$

(e) $y = x^2 + 4$, $y = 2x^2$, from $x = -2$ to $x = 4$.



$x^2+4 = 2x^2$

$4 = x^2$

$\pm 2 = x$

Region 1:

$A = \int_{-2}^{2} (x^2+4) - (2x^2)$

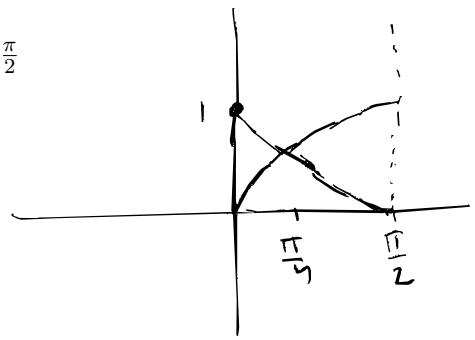
$= \int_{-2}^{2} 4 - x^2 \, dx$

Region 2:

$A = \int_{2}^{4} (2x^2) - (x^2+4)$ 2 areas

$= \int_{2}^{4} x^2 - 4 \, dx$

(f) $y = 1 - \sin x$, $y = \sin x$ from $x = 0$ to $x = \frac{\pi}{2}$



$A = \int_{0}^{\pi/6} (1-\sin x) - (\sin x)$

$= \int_{0}^{\pi/6} 1 - 2\sin x \, dx$

$1 - \sin x = \sin x$

$1 = 2\sin x$

$\sin x = \frac{1}{2}$

$x = \frac{\pi}{6}$

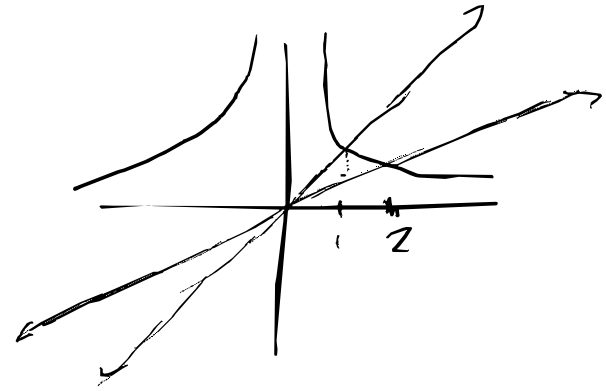
$A_2 = \int_{\pi/6}^{\pi/2} \sin x - (1-\sin x) \, dx$

$= \int_{\pi/6}^{\pi/2} 2\sin x - 1 \, dx$

4. Set up the integral(s) to find the area bounded by the curves $y = 1/x^2$, $y = x$, and $y = \frac{1}{8}x$.

$A_1 = \int_0^1 x - \frac{1}{8}x \, dx$

$A_2 = \int_0^2 \left(\frac{1}{x^2}\right) - \frac{1}{8}x \, dx$

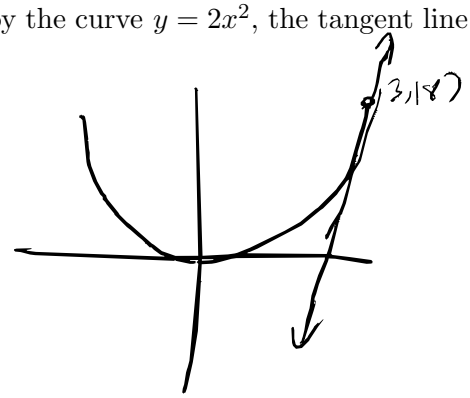


$\frac{1}{x^2} = \frac{1}{4}x$ $\quad$ $\frac{1}{x^2} = x$

$8 = x^3$ $\quad$ $1 = x^3$

$x = 2$ $\quad$ $x = 1$

5. Set up the integral(s) to find the area of the region bounded by the curve $y = 2x^2$, the tangent line to this curve at the point $(3, 18)$, and the x-axis.



$y' = 4x$

$y'(3) = 12$

$18 = (3)(12) + c$

$18 = 36 + c$

$-18 = c$

$y = 12x - 18$

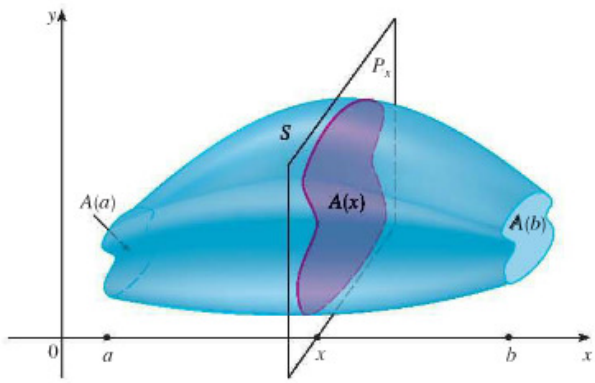
$x = \pm\sqrt{\frac{y}{2}}$

$\frac{1}{12}y + \frac{18}{12} = x$

$A = \int_0^1 \left(\frac{1}{12}y + \frac{3}{2}\right) - \left(\sqrt{\frac{y}{2}}\right) dy$

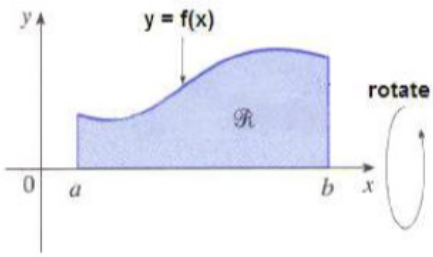
6.2 Volumes

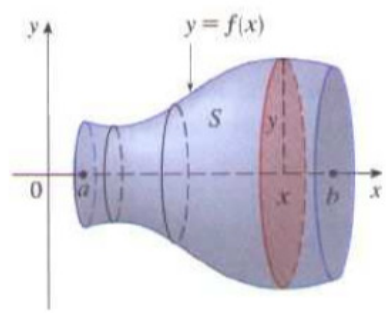
General Principle:



Disks

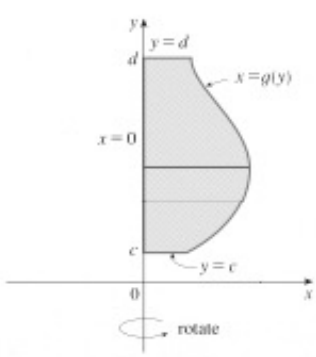
About x-axis:

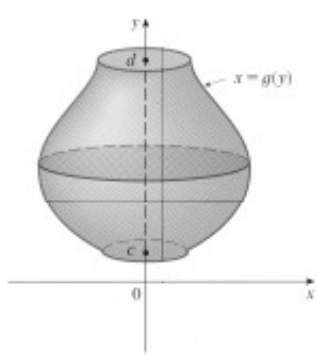




$A = \pi r_i^2 \qquad V = \pi (r_i)^2 dx \qquad V = \int_A^b \pi (R_i)^2 dx$

About y-axis:

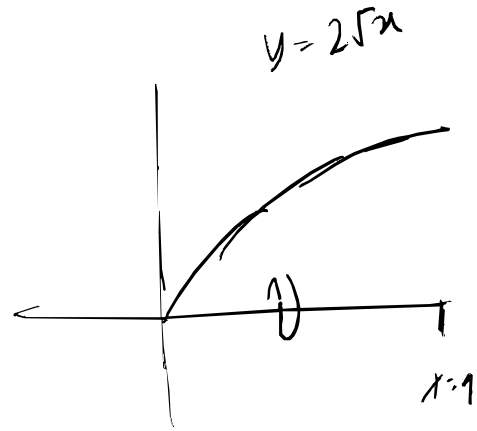




1. Find the volumes of the solids found by rotating the regions bounded by the given curves about the specified axis/line.

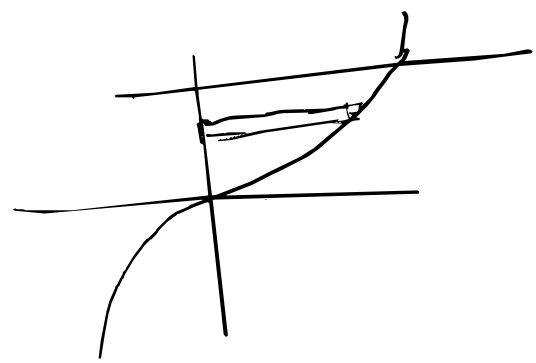
   (a) $y = 2\sqrt{x},\ y = 0,\ x = 9$ about the x-axis

$$V = \int_0^9 \pi (2\sqrt{x})^2 \, dx$$



   (b) $y = x^3,\ y = 8,\ x = 0$ about the y-axis

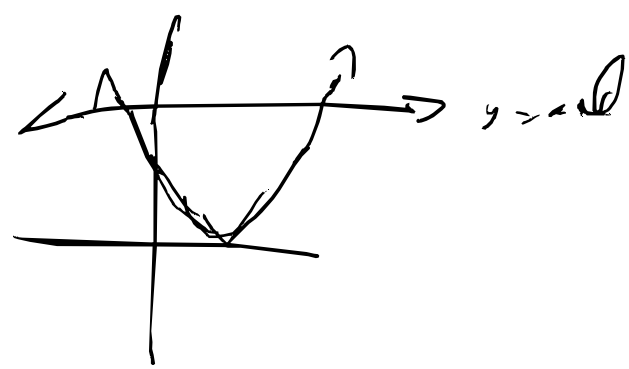
$$\int_0^8 \pi (\sqrt[3]{y})^2 \, dy$$



   (c) $y = (x-2)^2,\ y = 9$ about the line $y = 9$

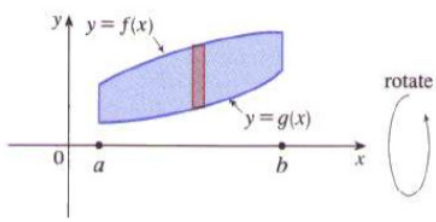
$$R_i = 9 - (x-2)^2$$

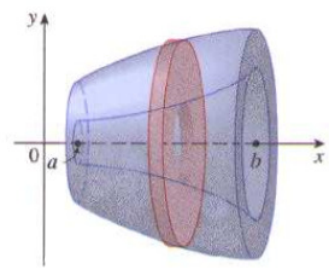
$$V = \int_0^9 \pi \left[9 - (x-2)^2 \right]^2 dx$$



(d) $x = (y-1)^2 + 2,\ x = 6$ about the line $x = 6$

<u>Washers</u>





For examples 2-4, find the volumes of the solids found by rotating the regions bounded the given curves about the specified axis/line.

2. (a) $y = x^2 + 2,\ y = 3x$ about the x-axis

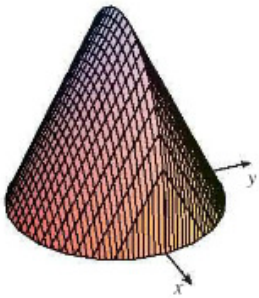
(b) $y = x^2 + 2,\ y = 3x$ (Same region) about the y-axis

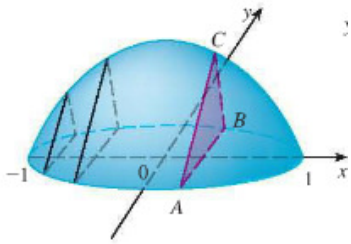
3. (a) $x = y^2,\ y = x^3$ about the line $y = 3$

(b) Same region about the line $x = -2$

Volumes by Cross-Sections

Example where base is a circle and cross-sections perpendicular to the x-axis are equilateral triangles:

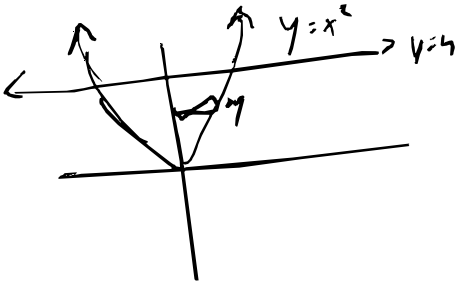




(a) The solid

We'll do a problem like this in the third example below. First, we'll do a few "easier" ones.

4. Find the volume of the solid S whose base is the region bounded by the curves $y = x^2$ and $y = 4$. Cross sections perpendicular to the y-axis are squares.

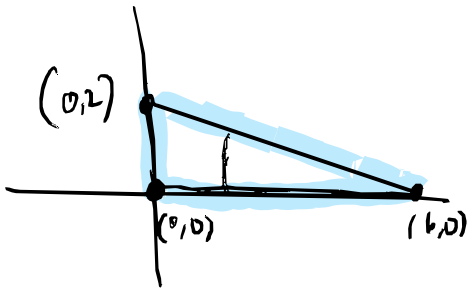


$x = \pm\sqrt{y}$

$$V = \int_0^4 (2\sqrt{y})^2 \, dy$$

$$V = \int_0^4 4y \, dy$$

5. Find the volume of the solid S whose base is the triangular region with endpoints $(0,0), (6,0)$, and $(0,2)$. Cross-sections perpendicular to the x-axis are semicircles.



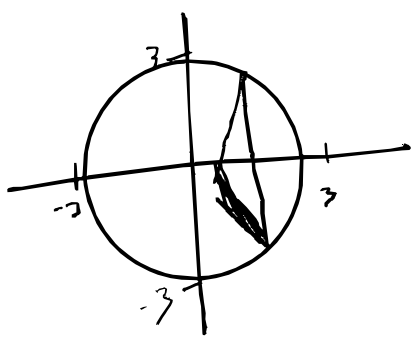
$y = -\frac{1}{3}x + 2 \quad = 0$

$r = -\frac{1}{6}x + 1$

$\frac{\pi r^2}{2}$

$V = \frac{\pi}{2}\int_0^6 \left(-\frac{1}{6}x+1\right)^2 \, dx$

6. Find the volume of the solid S whose base is the circle $x^2 + y^2 = 9$. Cross sections perpendicular to the x-axis are equilateral triangles.



$y^2 = 9 - x^2$

$\frac{1}{2}bh$

$y = \pm\sqrt{9-x^2}$

bottom $= 2\sqrt{9-x^2}$

$A = \frac{\sqrt{3}}{4}x^2$

$A = \sqrt{3}(9-x^2)$

$V = \int_{-3}^{3} \sqrt{3}(9-x^2)\,dx$